

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

A university has a database that contains information about students, courses, and enrollments. The students have attributes such as StudentID (primary key), Name, and Age. The courses have attributes like CourseID (primary key), CourseName, and Credits. An enrollment connects students to courses and includes attributes EnrollmentID (primary key), Grade, and Semester. There are many students for each course (many-to-many relationship), and a student can be enrolled in multiple courses.

- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider the following relation schema: BookInventory (ISBN, Title, AuthorID, PublisherID, Price) with the functional dependencies $F = \{\{ISBN \rightarrow Title, ISBN \rightarrow AuthorID, AuthorID \rightarrow PublisherID, AuthorID \rightarrow Price\}\}$. Analyze this schema for normalization.

- a) Draw the functional dependency diagram. (1)
- b) Use the attribute closure and identify the primary key for the given relation. (1)
- c) If we insert a row $\langle 978-3-16-148410-0, 'Database Systems', null, 123, 59.99 \rangle$ into the given relation, what is the update anomaly that is violated? Give reasons. (2)
- d) Assume that the relation has a row, $\langle 978-0-12-345678-9, 'Data Science Fundamentals', 456, 789, 49.99 \rangle$. If we remove the author with ID 456, what is the possible update anomaly that can be violated? Give reasons. (2)
- e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)
- f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)
- g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)
- h) Are the 3NF relations derived in part (e) in BCNF? If not bring the relations to BCNF. (2)

Question Q3 (25 marks)

A retail company is developing a comprehensive database system to manage its product inventory and sales data. The database structure defined includes tables for products, sales transactions, customers, and staff members. Ensure you address the various SQL commands and roles that will be implemented for managing the database effectively.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2)
- b) There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment given below? Briefly explain when this type of statement will be used. (2)
- c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2)
- d) Briefly explain how authentication and authorization are achieved in SQL Server. (3)
- e) A retail company has assigned various roles for its database. Mark is the lead database administrator responsible for overseeing the database system's management. Write a T-SQL statement to create a login for Mark with Windows authentication. (3)
- f) Provide Mark with the necessary permissions to perform all administrative operations within the server through a fixed server role. (3)
- g) Assuming Laura's username is 'laura.l', write T-SQL statements to assign her the responsibility of managing the sales transactions table using a user-defined role. (4)
- h) Assuming Alex's login name is 'alex.a', write T-SQL statements to grant him permission to create and modify product tables. (3)
- i) Assuming Jessica's login name is 'jessica.j', write T-SQL statements to allow her to execute data entry tasks on the product and sales transaction tables. (3)

Question Q4 (40 marks)

A university maintains records of its courses, students, and professors. Each course has a unique course code, a title, and is taught by one or more professors. Each student is identified by a student ID, has a name, and can enroll in multiple courses. Professors are identified by their faculty ID and have names as well. Draw an Entity-Relationship Diagram to represent the above scenario.

- a) Identify the main entities involved in the given scenario. (5)

- b) List the attributes for each entity identified. (10)
- c) Determine and state the relationships between the entities, including cardinalities. (10)
- d) Draw an ER diagram to represent the entities and relationships from the scenario. (10)
- e) Convert the ER diagram into a relational schema mapping each entity to a table structure. (5)